

# Justin Wu

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## EDUCATION

### Carnegie Mellon University

*Bachelor of Science in Artificial Intelligence*

*GPA: 4.0/4.0*

Pittsburgh, PA

*August 2026 – May 2029*

## EXPERIENCE

### Research Assistant

June 2024 – August 2026

*Nathanson Lab, Perelman School of Medicine, University of Pennsylvania*

*Philadelphia, PA*

- Conducted association testing on approximately 500K single-nucleotide polymorphisms (SNPs) across thousands of samples in a genome-wide association study (GWAS) meta-analysis of polygenic risk scores and cancer relapse in testicular germ cell tumors, producing Manhattan and QQ plots to assess genome-wide significance
- Engineered a Bash/Python pipeline to identify proxy SNPs in linkage disequilibrium ( $r^2 \geq 0.8$ ) with 100+ lead GWAS hits across 500 kb windows, using LocusZoom for regional LD visualization to enable downstream fine-mapping
- Harmonized SNP naming and genome builds across the Penn Medicine BioBank (PMBB), Gencove, and Discovery cohorts, correcting an outdated marker list and tripling the size of the shared replication set

## PROJECTS

### Apex

June 2025 – June 2026

*3D F1 Race Replay Engine | Three.js, Vite, FastF1*

- Built an interactive 3D replay engine visualizing ~140 Formula 1 races since 2020 from real FastF1 telemetry, rendering ~250K data points per race at a sustained 60 FPS
- Applied Kalman filtering to smooth noisy telemetry, cutting car jitter by 50%, and physics-informed splines to interpolate car paths without overlap between cars
- Implemented multiple camera modes (free orbit, follow car, TV chase/action cams) and variable-speed playback (0.5x–64x) synchronized with telemetry

## PUBLICATIONS & PRESENTATIONS

### Enhancing Genetic Ancestry Inference through TDA and Clustering

July 2026

*Wu, J., & Yang, T. — Poster, Intelligent Systems for Molecular Biology (ISMB) Conference*

- Paired topological features with unsupervised clustering to improve ancestry inference in admixed populations, reaching 96% accuracy on the admixed American (AMR) superpopulation versus 73% for PCA with k-means
- Applied persistent homology to 1000 Genomes Project SNP data from 2,504 individuals across 26 populations, generating persistence barcodes to capture population structure beyond what linear methods such as PCA reveal
- Maintained overall ancestry classification accuracy of 99.08% using topological data analysis (TDA) with Ward linkage clustering, on par with PCA with k-means (99%)

## TECHNICAL SKILLS

**Languages:** Java, Python, C/C++, JavaScript, HTML/CSS, R, Swift

**Developer Tools:** Git, Docker, VS Code, Visual Studio, PyCharm, Eclipse, Bash, npm

**Libraries:** pandas, NumPy, Matplotlib, SciPy, scikit-learn, seaborn, statsmodels, Biopython

## HONORS & AWARDS

- 2x American Invitational Mathematics Examination (AIME) Qualifier (2023, 2025)
- USA Biolympiad, Recognition of Academic Excellence — top 50 of 10000+ competitors (2025)
- MathWorks Math Modeling (M3) Challenge, Honorable Mention — top 25 of 800+ papers (2025)
- Future Business Leaders of America (FBLA) PA State Champion, Computer Problem Solving (2025)
- Pennsylvania Seal of Biliteracy, Spanish (2026)